

5. Introduction to Jenkins: What is Jenkins?, Installing Jenkins on Local or Cloud Environment, Configuring Jenkins for First Use.

Introduction to Jenkins

Jenkins is a popular open-source automation server used for continuous integration (CI) and continuous delivery (CD) in software development. It helps automate building, testing, and deploying applications, making development more efficient.

What is Jenkins?

Jenkins is a Java-based automation tool that enables developers to:

- Automate repetitive tasks in software development.
- Continuously integrate code changes and run tests.
- Deploy applications to different environments efficiently.
- Monitor and manage CI/CD pipelines with a web-based interface.

Key Features of Jenkins

- Extensible: Supports hundreds of plugins for integration with various tools.
- Distributed Architecture: Supports master-slave configurations for scalable builds.
- Easy to Use: Web-based interface with simple configurations.
- Supports Multiple Environments: Works on Windows, Linux, and Mac, as well as cloud-based platforms.

Installing Jenkins

Jenkins can be installed on local machines, on a cloud environment, or even in containers.

Jenkins Installation Steps on Ubuntu (Local System):

Step 1: Update System Packages

Open Terminal and run:

`sudo apt update`

Step 2: Install Java (Required for Jenkins)

Jenkins requires Java. Install OpenJDK:

`sudo apt install openjdk-17-jdk -y`

Check Java version:

`Java -version`

Step 3: Go to Official Jenkins Website

www.jenkins.io

Step 4: Go to Download Jenkins 2.541.2 LTS for:

`Click on debian/ubuntu`

`Copy commands from “Long Term Support release” and paste on terminal for installation.`

Step 5: Start Jenkins Service

`sudo systemctl start Jenkins`

Step 6: Enable Jenkins at boot

`sudo systemctl enable Jenkins`

Step 7: Check status

`sudo systemctl status Jenkins` (You should see: active (running))

Step 8: Open Jenkins in Browser

Open browser and go to:

<http://localhost:8080>

Step 9: Get Initial Admin Password

Run:

sudo cat /var/lib/jenkins/secrets/initialAdminPassword

(Copy the password and paste it in Jenkins browser setup.)

Step 10: Complete Setup

- **Click Install Suggested Plugins**
- **Create Admin User**
- **Click Save and Finish**

Installing Jenkins on the Cloud

Jenkins can be installed on cloud platforms like AWS, Azure, and Google Cloud.

Using AWS EC2:

- Launch an EC2 instance with Ubuntu.
- Connect to the instance via SSH.
- Install Java and Jenkins using the Linux commands above.
- Configure security groups to allow access to port 8080.
- Access Jenkins using the public IP of the instance (http://<EC2-Public-IP>:8080).